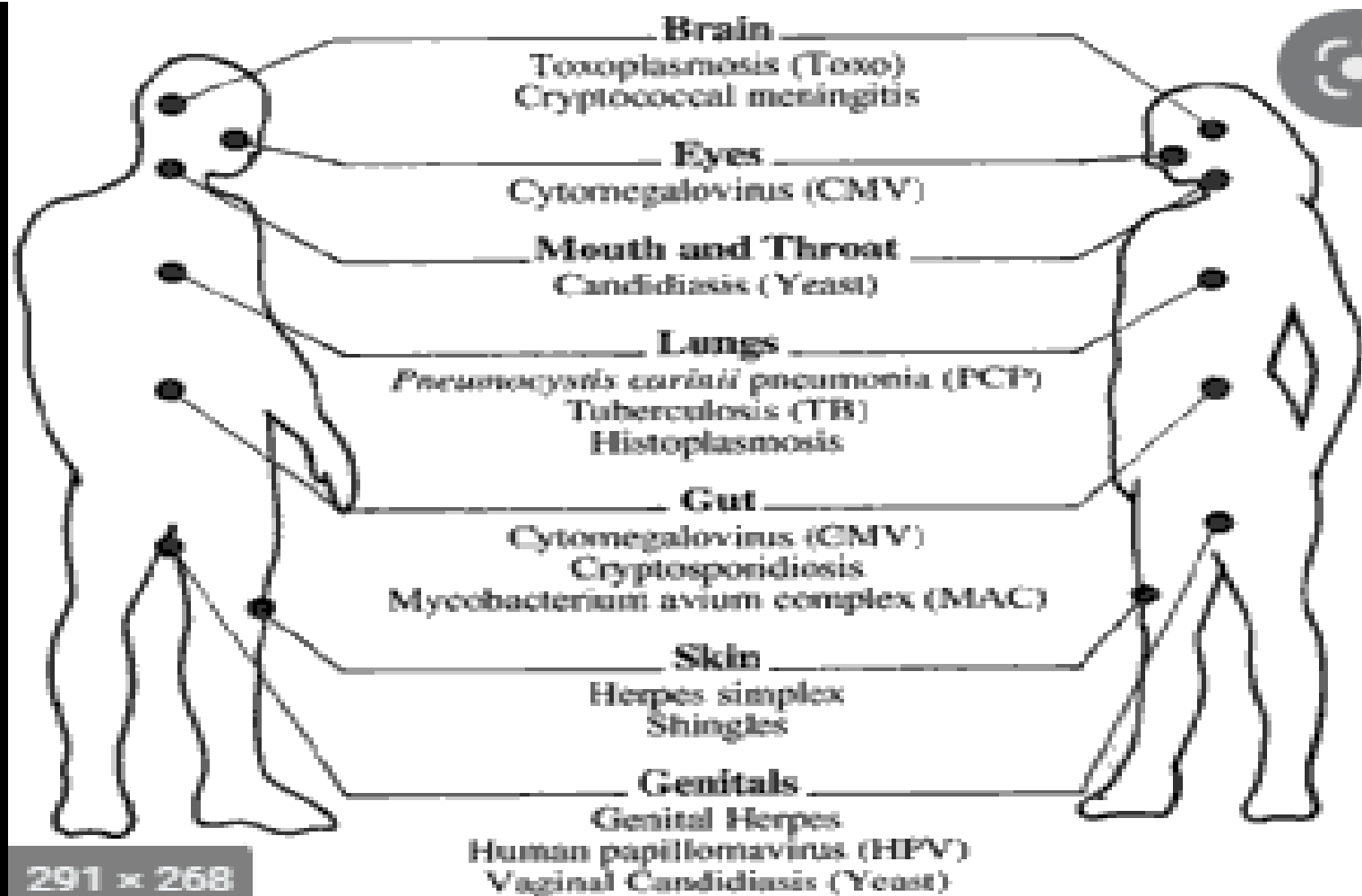


Opportunistic Infections

Opportunistic Infections

- Opportunistic infections are infections that occur more often or are more severe in persons with a weakened immune system.
- This can be generally seen in advanced stages of HIV or AIDs with CD4 < 200 cells μ L.

Common Opportunistic Infections



Candidiasis

- One of the most common opportunistic infections that occur in immunocompromised patients
- Oropharyngeal and esophageal candidiasis are common in patients with HIV infection

- The advent of antiretroviral therapy (ART) has led to a dramatic decline in the prevalence of oropharyngeal and esophageal candidiasis.

Oropharyngeal Candidiasis

- painless, creamy white, plaque-like lesions that can occur on the buccal surface, hard or soft palate, oropharyngeal mucosa, or tongue surface.
- can be easily scraped off with a tongue depressor or other instrument.



Esophageal Candidiasis

- PLHIV with oropharyngeal candidiasis can also have esophageal involvement
- retrosternal burning pain or discomfort
- odynophagia (pain on swallowing)

Treatment

Oropharyngeal

- Fluconazole 100mg once daily. x 1 week
- Nystatin 400,000-600,00IU qid

Esophageal

- Fluconazole 200-300mg once daily x 2-3 weeks

Pneumocystis pneumonia (PCP)
or
Pneumocystis jirovecii pneumonia
(PJP)

- Caused by the organism *Pneumocystis jirovecii*
- Symptoms:
 - Progressive shortness of breath
 - Non productive cough
 - Fever

- Auscultatory physical findings
 - often limited
 - not in keeping with the level of hypoxia at presentation.

- Chest X-Ray
 - may show bilateral, diffuse, interstitial infiltration involving all portions of the lung

PCP Treatment

Type of Treatment	Medication	Dosage
Prophylaxis (CD4 <350 cells/ μ l)	TMP/SMX or Dapsone	1 D/S tablet po od /100mg po od
Treatment	TMP/SMX	2 D/S tablets po tds x 21days Or 15-20mg/kg/day q6h p.o./IV x 21 days
Alternatives	Clindamycin	600-900mg IV q6h. x 21 days, 300-450mg p.o. q6h. x 21 days

- For PCP, the optimal time of ART commencement is within 10-14 days of therapy, once PCP therapy is shown to be tolerated.

Tuberculosis

- Tuberculosis (TB) is a disease caused by a bacterium called *Mycobacterium tuberculosis*.
- TB usually affects the lungs, but it can affect any part of the body, including the kidneys, spine, or brain.
- If not treated, TB can cause death.

Clinical presentation varies. There are Two main types of TB;

1. Pulmonary TB-this involves an individual's lung tissue.
2. Extrapulmonary TB-this involves other organs.

Clinical Presentation

- **constitutional symptoms: fever, night sweats, weight loss, cough >2-3 weeks**
- haemoptysis.
- pulmonary infiltrates in upper lobe with possible cavitation
- Lower CD4 counts : Extra-pulmonary and atypical pulmonary presentations become more common

- Extra-pulmonary infection
 - Lymph nodes – lymphadenitis
 - Brain – Meningitis, tuberculoma
 - Bone – typically spinal involvement (Pott's Disease)
 - Adrenal gland – adrenal insufficiency

Diagnosis:

- History and examination
- Investigations
 - Mantoux test (>5mm in the HIV co-infected patient)
 - Chest X-ray
 - Sputum analysis for acid-fast bacilli, culture and sensitivity.

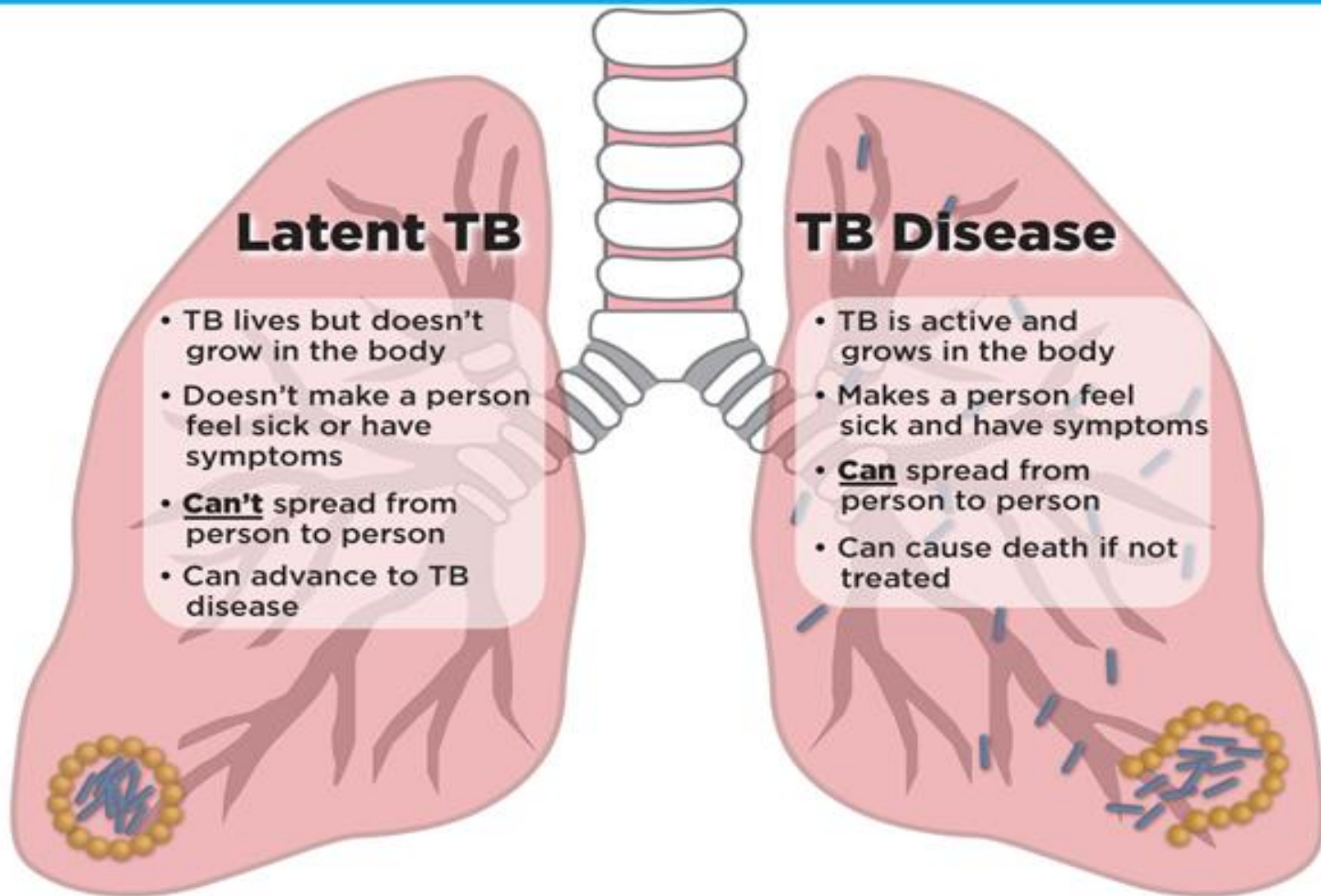
Asymptomatic cases with a positive Mantoux test >5mm and a negative chest x-ray should be diagnosed as Latent TB

Latent TB

- TB lives but doesn't grow in the body
- Doesn't make a person feel sick or have symptoms
- **Can't** spread from person to person
- Can advance to TB disease

TB Disease

- TB is active and grows in the body
- Makes a person feel sick and have symptoms
- **Can** spread from person to person
- Can cause death if not treated



- Requires admission to hospital
- Active pulmonary disease should be treated for a total of 6 months with a rifampin containing regimen
 - Induction: 2 months with 4 active agents (rifampin, isoniazid, ethambutol and pyrazinamide)
 - Maintenance: 4 months with 2 active agents (rifampin and isoniazid)

- For extra-pulmonary disease, treatment should be extended to between 9-12 months depending on the site of infection.
- Latent TB should be treated with isoniazid 300mg with vitamin B6 50mg daily for 6 months.

ART commencement in the face of active TB:
Tolerability of anti-TB therapy must first be demonstrated prior to ART commencement

- IRIS is of particular concern with early initiation of ART
- CD4 < 50 cells/ml consider ART after 2 weeks of anti-TB therapy
- CD4 > 50 cells/ml ART can be delayed up to 8 weeks of anti-TB therapy.
- All patients should be offered TMP/SMX prophylaxis.

Herpes Simplex Virus

- Infections with human herpes simplex virus type 1 (HSV-1) and type 2 (HSV-2) are common
- Grouped vesicular lesions, usually in the oral and anogenital region
- Large chronic erosions refractory to treatment can be considered as AIDS defining

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- Orolabial herpes
 - commonly known as cold sores or fever blisters
 - most common manifestation of HSV-1 infection
 - sensory prodrome in the affected area, rapidly followed by lesions on lips and oral mucosa that evolve in stages from papule to vesicle, ulcer, and crust.



- Genital herpes
 - most common manifestation of HSV-2 infection.
 - ulcerative lesions are usually the only stage observed on mucosal surfaces
 - vesicles are commonly seen on skin on or around the genitals (e.g., the penile shaft, mon pubis, thighs)
 - local symptoms might include a sensory prodrome consisting of pain and pruritus.

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■ Treatment

- Acyclovir 400mg q8h for 7-10 days
- Valacyclovir 1gm twice daily for 7-10 days.

Chronic suppressive therapy

- Valacyclovir 500mg b.d. or acyclovir 400mg b.d.

Herpes Zoster

- Also known as shingles
- Can occur in adults with HIV at any CD4 count, but with CD4 counts but with CD4 counts <200 the risk is higher
- HIV viremia is associated with an increased risk for incident herpes zoster.

Symptoms

- Painful vesicular lesions
- This can have multiple dermatomal involvement
- Often preceded by prodromal pain

- Approximately 10% to 15% of people with HIV report post-herpetic neuralgia as a complication following herpes zoster.

Herpes Zoster Treatment

Type of Treatment	Medication
Acute	Acyclovir 800mg po 5x daily for 7- 10 days Or Valacyclovir 1g p.o. t.i.d. x 7-10 day
Severe	Acyclovir 30mg/kg IV daily x 7-10 days

Cryptococcal Meningitis

- Cryptococcal meningitis is a serious infection of the lining of the brain and spinal cord caused by this fungus.
- *Cryptococcus* is a common fungus that is often found in soil and bird droppings.

- Cryptococcal Meningitis may be seen in patients with low CD4 counts of <100 cells/ μ L
- **Symptoms: Fever, dull headache, malaise, blurred vision, altered personality, altered mental status, nausea, vomiting**
- Diagnosed by doing a Lumbar puncture on the patient and examination of the Cerebrospinal fluid by lab technicians.

Cryptococcal Meningitis Treatment

Type of Treatment	Medication	Side Effects
Initial Phase	Amphotericin B + Flucytosine Or Amphotericin B + Fluconazole Or Fluconazole	Amphotericin B may cause side effects such as nausea vomiting/ upset stomach/ Rash or swelling at injection site other serious side effects include irregular heart beat or decrease urination
Consolidation Phase	Fluconazole 800 mg po od x 8 weeks	Gastrointestinal Nausea/ Vomiting heart burn Skin rashes
Maintenance	Fluconazole 200mg daily until CD4 counts > 200cells/mm ³ for at least 1 year.	

- ART initiation should be deferred until there is evidence of a sustained clinical response to antifungal therapy
- After 4 weeks of induction and consolidation treatment with amphotericin and fluconazole or 4-6 weeks of treatment with high dose oral fluconazole induction and consolidation.

Toxoplasmosis

- Most common CNS infection in patients with AIDS (CD4 counts < 200) who are not receiving appropriate prophylaxis
- This infection has a worldwide distribution and is caused by the intracellular protozoan parasite, *Toxoplasma gondii*.

- Symptoms
 - +/- fever
 - headache
 - altered mental status e.g. confusion
 - lethargy
 - delusional behaviour
 - cognitive impairment in 60% of patients
 - seizures

- Investigations
 - IgG positive serology
 - CT scan: shows multiple ring enhanced lesions of toxoplasmosis, however nothing may be seen
 - If no treatment response after initial 2 weeks of therapy, may consider brain biopsy

Toxoplasmosis Treatment

Type of Treatment	Medication
Prophylaxis (CD4 <200 cells/ μ l)	TMP/SMX Or Dapsone 1 D/S tablet po od 50mg po od + Pyrimethamine 50mg/week
Treatment	Pyrimethamine 200mg loading dose then 50mg (<60kg weight) daily 75mg (>60kg weight) daily + folinic acid 10 -25mg daily
Alternatives	Clindamycin Or TMP/SMX 600- 900mg IV q6h. x 21 days, 10- 15mg/kg/d divided q6- 8h X 6weeks

Mycobacterium Avium Complex

- MAC is a group of bacteria related to tuberculosis.
- MAC usually causes problems at CD4 cell counts < 50
- Requires referral to a specialist centre

- It can infect one part of the body, like the lungs, bones, or intestines. This is localized infection.
- It can also spread and cause disease throughout the body. This is called disseminated infection.

■ Symptoms

- May present as fever, malaise, weight loss, anaemia, neutropaenia
- Chronic diarrhoea and abdominal pain
- Extra-biliary obstructive jaundice

Mycobacterium Avium Complex Treatment

Type of Treatment	Medication
Acute	Clarithromycin + Ethambutol +Rifabutin 500mg bd 15mg/kg/day 300mg od For a minimal of 1 year
Alternative	Azithromycin replaces Clarithromycin above 500mg od

- MAC prophylaxis
- CD4 count < 100/mm³
- Azithromycin 1250mg weekly OR
Clarithromycin 500mg b.d

Immune Reconstitution Inflammatory Syndrome (IRIS)

- Worsening of signs and symptoms due to known infections or the development of disease due to occult infections
- Results from an excessive inflammatory response by a re-invigorated immune system following the initiation of anti-retroviral therapy

- Clinical manifestations vary depending on the underlying opportunistic infection, generally more severe symptoms than those expected from the responsible opportunistic infection.

- Presentation can occur anywhere between 6 weeks to 6 months after commencement of ART.
- Treatment generally involves the use of NSAIDs or steroids.

- For opportunistic infections with a higher risk for immune reconstitution syndrome (Cryptococcus, non-tuberculous mycobacterial infections), initial phases of therapy for the opportunistic infection should be completed prior to ART commencement

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Thank
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